

Cost Function Intuition

The primary goal of a learning algorithm is to find the parameters (w and b) that make the cost function $J(w, b)$ as small as possible. This process is formally written as finding the **minimum** of the cost function:

$$\min_{w,b} J(w, b)$$

A Simplified Model for Visualization

To make it easier to visualize how the cost function works, we can temporarily simplify the linear model by removing the b parameter (or setting $b=0$).

- **Original Model:** $f_{w,b}(x) = wx + b$
- **Simplified Model:** $f_w(x) = wx$
- **Simplified Cost Function:**
$$J(w) = \frac{1}{2m} \sum_{i=1}^m (f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} - y^{(i)})^2$$

With this simplification, the model is a line that is forced to pass through the origin (0,0), and we only need to find the best value for a single parameter, w .

Relating the Model to the Cost Function

It's crucial to understand the relationship between the model plot and the cost function plot.

- **Model Plot (f vs. x):** This shows the relationship between the input x and the output y . A specific value of w defines a specific line on this plot.
- **Cost Function Plot (J vs. w):** This shows how "good" a particular parameter w is. The horizontal axis is the parameter w , and the vertical axis is the cost $J(w)$ associated with that w .

For every possible line on the model plot, there is a corresponding point on the cost function plot.

How the Cost $J(w)$ Changes with w

Let's trace how the cost changes for different values of w using a simple training set: $(1, 1)$, $(2, 2)$, $(3, 3)$.

1. If $w = 1$:

- The model is $f(x) = 1x$, which is a perfect fit for the data.
- The error $(\hat{y} - y)$ is 0 for every data point.

- The cost $J(1) = 0$. This is the lowest possible cost.

2. If $w = 0.5$:

- The model is $f(x) = 0.5x$. This line is below the data points.
- There is a vertical distance (an error) between each data point and the line.
- The cost $J(0.5)$ is a positive number (calculated as ≈ 0.58).

3. If $w = 0$:

- The model is $f(x) = 0x$, a flat horizontal line.
- The errors are even larger.
- The cost $J(0)$ is higher still (calculated as ≈ 2.33).

By trying many different values for w and plotting the resulting cost $J(w)$, we can trace out the shape of the cost function. For linear regression, this shape is a **bowl-shaped curve** (a parabola).

The **goal of linear regression** is to find the value of w (and b in the full model) that is at the very bottom of this bowl, as this corresponds to the model with the lowest possible error.